Software Testing Test Cases

Equivalence Partitioning (EP) & Boundary Value Analysis (BVA)

1. Write equivalence partitioning (EP) test cases for the phone number input field in a Zomato-style food order form, where the field must accept exactly 10 digits and no letters or special characters.

EQUIVALENCE CLASSES

Partition	Type	Example
Exactly 10 digits	**Valid**	9876543210
Less than 10 digits	**Invalid**	987654321
More than 10 digits	**Invalid**	98765432101
Contains letters	**Invalid**	98765abcde
Contains special characters	**Invalid**	98765@3210
Empty field	**Invalid**	""

TEST CASES

TC ID	Input	Expected Result
TC01	9876543210	**Accepted**
TC02	1234567890	**Accepted**
TC03	987654321	**Rejected**
TC04	98765432101	**Rejected**
TC05	98765abcde	**Rejected**
TC06	98765@3210	**Rejected**
TC07	Empty	**Rejected**

> **2. For a Flipkart-style sign-up form with a password field that must be 8-16 characters long, write boundary value analysis (BVA) test cases to check the minimum, just above minimum, maximum, and just above maximum password lengths.**

Hint: *Consider test cases for 7, 8, 9, 15, 16, and 17 characters.*

BOUNDARY VALUES

- Minimum - 1 = 7
- Minimum = 8
- Minimum + 1 = 9
- Maximum - 1 = 15
- Maximum = 16
- Maximum + 1 = 17

TEST CASES

TC ID	Length	Sample Password	Expected Result
TC01	7	Pass123	Rejected
TC02	8	Pass1234	Accepted
TC03	9	Pass12345	Accepted
TC04	15	Password1234567	Accepted
TC05	16	Password12345678	Accepted
TC06	17	Password123456789	Rejected

> **3. Given an Instagram-like post creation form with a caption input that allows up to 2200 characters, use equivalence partitioning to identify valid and invalid input ranges and write at least 3 test cases for each partition.**

PARTITIONS

Partition	Type
0–2200 characters	Valid
More than 2200 characters	Invalid

VALID TEST CASES

TC ID	Input Length	Expected Result
TC01	0 characters	Accepted
TC02	100 characters	Accepted
TC03	2200 characters	Accepted

INVALID TEST CASES

TC ID	Input Length	Expected Result
TC04	2201 characters	Rejected
TC05	2500 characters	Rejected
TC06	3000 characters	Rejected

> **4. For a BookMyShow movie ticket booking form with a 'number of tickets' field (allowed: 1 to 10), write BVA test cases to ensure the field correctly handles edge values.**

BOUNDARY VALUES

- Min - 1 = 0
- Min = 1
- Min + 1 = 2
- Max - 1 = 9
- Max = 10
- Max + 1 = 11

TEST CASES

TC ID	Tickets	Expected Result
TC01	0	Rejected
TC02	1	Accepted
TC03	2	Accepted
TC04	9	Accepted
TC05	10	Accepted
TC06	11	Rejected

5. Pick any app you use daily (such as Paytm, Myntra, or IRCTC) and select one form field (e.g., amount, pincode, age). Write both equivalence partitioning and boundary value analysis test cases for that field, explaining your reasoning for each.

Selected App: Paytm (or similar payment app)
Field Requirement: Amount must be between ₹1 and ₹10,000

A. EQUIVALENCE PARTITIONING (EP)

Partitions

Partition	Type
₹1 – ₹10,000	Valid
Less than ₹1	Invalid
Greater than ₹10,000	Invalid
Letters / Special Characters	Invalid

Test Cases

TC ID	Input	Expected Result
TC01	₹500	Accepted
TC02	₹1	Accepted
TC03	₹10,000	Accepted
TC04	₹0	Rejected
TC05	₹-50	Rejected
TC06	₹10,001	Rejected
TC07	abc	Rejected
TC08	@500	Rejected

Reasoning

- One test case is selected from each valid and invalid partition.
- If one value works in a partition, other values in the same partition are assumed to behave similarly.

B. BOUNDARY VALUE ANALYSIS (BVA)

Boundary Values

- Min - 1 = 0
- Min = 1
- Min + 1 = 2
- Max - 1 = 9999
- Max = 10000
- Max + 1 = 10001

Test Cases

TC ID	Amount	Expected Result
TC01	₹0	Rejected
TC02	₹1	Accepted
TC03	₹2	Accepted
TC04	₹9,999	Accepted
TC05	₹10,000	Accepted
TC06	₹10,001	Rejected

Reasoning

- Boundary values are most likely to contain defects, so testing values just below, at, and just above the limits helps identify errors efficiently.